ICAO'S INVOLVEMENT IN OUTER SPACE AFFAIRS – A NEED FOR CLOSER SCRUTINY?

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I. INTRODUCTION

The question as to whether the International Civil Aviation Organization (ICAO) should be involved in the law making process pertaining to outer space affairs has already been addressed by scholars and professionals. Dr. Assad Kotaite, President of the ICAO Council, drew the inextricable link between space technology and civil aviation when he observed that space technology has made far reaching contributions to the betterment of the human condition, and civil aviation was one sector where such technological benefit was evident.[1] It is irrefragable that space science and technology will play a critical role in the future development of civil aviation. In the words of Dr. Kotaite:

> Laid out on the drawing boards of aircraft manufacturers and futurists are spacecraft that one day will carry passengers into the upper airspace and eventually into outer space. When that day comes, and it may not be that far away, real issues will need to be addressed by government regulators.... The idea of adopting ICAO as a model, or expanding the mandate of ICAO to encompass outer space has merit.[2]

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[1] Assad Kotaite, *Formal Regulatory Framework Needed to Govern Expanding Operations in Outer Space*, 55 ICAO J. 5 (2000). Dr. Kotaite observed that ICAO has been actively promoting the use of space technology in aviation since 1972 when the Future Air Navigation System (FANS), the precursor to the Communications, Navigation, Surveillance/Air Traffic Management System (CNS/ATM), was discussed at ICAO's 7th Air Navigation Conference. *Id. See also generally*, Ruwantissa I.R. Abeyratne, *Legal and Regulatory Issues in International Aviation*, *in* SPACE TECHNOLOGY AND AIR NAVIGATION, 260-269 (Transnational Publishers Inc., New York, 1996).

[2] Kotaite, *supra* note 1, at 5.

Wassenbergh made a similar observation in 1999 when he said that, in the context of launch vehicles, ICAO's mandate could well be expanded to include the drafting of "minimal traffic safety, security and environmental rules" for launch vehicles.[3] The inextricable link between the communications, navigation, survellience/air traffic management (CNS/ATM) system and satellite technology brings to bear the compelling importance of the Aeronautics Telecommunications Network (ATN) for which ICAO has prolifically set standards in Annex 10 to the Convention on International Civil Aviation.[4] It is believed that ICAO's demonstrable competence in adopting 18 Annexes to the Chicago Convention, all of which contain Standards and Recommended Practices (SARPs) to regulate international civil aviation efficiently over the past 60 years, may well make it the global forum of nations which may be needed to achieve consensus in the management of outer space.[5] From a technical standpoint, ICAO's credibility is strengthened by the example set by the Organization in expertly migrating to the CNS/ATM system, thus achieving a seamless and global air traffic system.[6] A strong argument has already been made to separate the launching activity from outer space activity on the basis that space activities are essentially activities taking place in outer space and that the launching activity is a transportation activity occurring in the Earth's airspace.[7] Wassenbergh states:

> I am of the opinion that there is reason for separating the international legal regimes applicable to these activities, espe-

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[3] Henri Wassenberg, *Access of Private Entities to Airspace and Outer Space*, 24 ANNALS OF AIR & SPACE L. 311, 316 (1999).

[4] Convention on International Civil Aviation, Dec. 7, 1944, 15 UNTS 295, UKTS 8 [hereinafter Chicago Convention]. *See* ICAO Doc 7300/8 (8th ed., Sept. 2000), *available at* http://www.icao.int/cgi/goto_m.pl?/icao/en/download.htm#Docs (last viewed Oct. 9, 2004).

[5] Kotaite, *supra* note 1, at 5.

[6] Jack Howell, *Evolutionary Approach to Transition Now Focussed on Detailed Vision of How to Exploit Technologies*, 55 ICAO J. 6 (Sept. 2000).

[7] Henri A. Wassenbergh, *A Launch and a Space Transportation Law, Separate From Outer Space?*, 21 AIR AND SPACE L. 1, 28 (1996).

> cially where it concerns activities deployed by non-governmental entities.[8]

He suggests that a clear distinction must be made between activities in outer space and transportation leading up to the journey in outer space. The latter would include the launch of the vehicle from the ground which would involve traversing airspace above the territory of a State or States. Wassenbergh's opinion cannot be ignored as it states an incontrovertible fact. However, the real issue lies in the determination of the common elements of air and space travel and the building of a common interoperable and international legislative structure that could encompass both the transitional process from air space to outer space and the application of the philosophy of each legal system to the other. Inextricably linked to this ultimate goal is the need to initially determine the synergies between air transport and space transport, both from technical and legal perspectives.

II. ELEMENTS COMMON TO AIR LAW AND SPACE LAW

Air law came into being from a global regulatory perspective when 52 States signed the Chicago Convention on 7 December 1944.[9] Space law was launched in 1958 when the United Nations General Assembly adopted Resolution 1348.[10] The fundamental distinction between the legal systems is that air law is both more structured and all encompassing as its regulatory aspects stems from one international treaty. Space law on the other hand has no all encompassing regulatory document nor an international organization responsible for its regulation in the nature of a specialized agency of the United Nations system. Instead, several international treaties address different issues at space law.[11] However, both systems have certain fundamental

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[8] *Id.*

[9] Chicago Convention, *supra* note 4.

[10] Question of the Peaceful Use of Outer Space, U.N. ICAO, 792nd Plenary Meeting, U.N. Doc. A/RES/1348 (1958), *available at* www.un.org.

[11] The basic space treaties are:

1. Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Jan.

commonalities which are hard to ignore. For one, both the Chicago Convention and A/RES/1348 refer to the needs of the people of the world, where the former states that the development of the techniques of air navigation and principles of air transport have to meet the needs of the peoples of the world while the latter recognizes the common interest of humankind in outer space.[12] Article 1 of the Chicago Convention embodies the principle of sovereignty of nations while the second preambular clause of A/RES/1348 makes reference to Article 2 (1) of the Charter of the United Nations which states that the United Nations is based on the sovereign equality of all its Members. The Preamble to the Chicago Convention refers to the future development of civil aviation benefiting humankind and a similar provision is contained in the fourth preambular clause of A/RES/1348 which expresses the desire of the General Assembly to promote energetically the fullest exploration and exploitation of outer space for the benefit of humankind.

Although A/RES/1348 considered the importance of the contribution that could be made by an appropriate international body for cooperation in the study of outer space for peaceful purposes, and indeed established the United Nations Commit-

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27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 (entered into force on Oct. 10. 1967);

2. Agreement on the Rescue of Astronauts, the Return of Astronauts, and the Return of Objects Launched into Outer Space, Apr. 22, 1968, 19 U.S.T. 7570, 672 U.N.T.S. 119;

3. Convention on International Liability for Damage Caused by Space Objects, Mar. 29, 1972, 24 U.S.T. 2389, 961 U.N.T.S. 187;

4. Convention on Registration of Objects Launched into Outer Space, Jan. 14 1975, 28 U.S.T. 695, 1023 U.N.T.S. 15; and

5. Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, Dec. 18 1979, 1362 U.N.T.S. 3.

Other treaties explicitly apply in part to activities in space but are not limited to such activities, e.g. Treaty Banning Nuclear Weapons Tests in the Atmosphere, in Outer Space and Under Water, Aug. 5, 1963, U.S.-U.S.S.R., 14 U.S.T. 1313; and Treaty on the Limitation of Anti-Ballistic Missile Systems, May 26, 1972, U.S.-U.S.S.R., 23 U.S.T. 3435. The bilateral arms limitation agreements between the United States and the Soviet Union may also qualify as international space law to the extent that they prohibit interference with "national technical means of verification," including satellite reconnaissance.

[12] *See* Chicago Convention, *supra* note 4, at Preamble and art. 44 (d).

tee on the Peaceful Uses of Outer Space (UNCOPUOS),[13] such a measure does not match up to the signal initiative of the Chicago Convention. Article 44 of that Convention established ICAO. It is a specialized agency of the United Nations which comprehensively provides guidelines for international civil aviation through the 18 Annexes to the Convention as well as various other Assembly and Council resolutions and policy. UNCOPUOS has neither law making nor adjudicative powers[14] whereas the Chicago Convention contains several usable provisions that are enforceable[15] and offers the ICAO Council effective enforcement powers that could be used to remedy inadequacies in the global air navigation system.[16] It has been acknowledged that ICAO possesses genuine rule making powers and authority on matters of navigation over the high seas and other ocean areas where there is freedom of overflight.[17]

Another commonality between air law and space law is the heavy reliance placed by both systems on the sovereign right of nations to carry out functions in aviation and outer space activities. As mentioned earlier, the express mention of sovereignty in Article 1 of the Chicago Convention manifests itself in a practical sense regularly. One of the latest examples is the Global Declaration adopted at the 5th Worldwide Air Transport Conference of ICAO in March 2003 which recognized the sovereign right of each State to charter its own path of liberalization of air

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[13] Committee on the Peaceful Uses of Outer Space, U.N. Doc. A/RES/1472A (Dec. 12, 1959) [hereinafter COPUOS].

[14] Ralph G. Steinhardt, *Outer Space, in* 2 UNITED NATIONS LEGAL ORDER 753, 757 (Oscar Schachter & Christopher C. Joyner eds. Cambridge Univ. Press 1995).

[15] Chicago Convention, *supra* note 4, at arts. 54(j)-(k) (giving ICAO Council authority to monitor conduct of Contracting States).

[16] *Id.* at art. 69. Article 69 of the Convention provides that if the Council is of the opinion that the airports or other air navigation facilities, including radio and meteorological services, of a Contracting State are not reasonably adequate for the safe, regular, efficient and economical operation of international air services, present or contemplated, the Council shall consult with the State directly concerned, and other States affected, with a view to finding means by which the situation may be remedied, and may make recommendations for that purpose. *Id.* No Contracting State shall be guilty of an infraction of the Convention if it fails to carry out these recommendations. *Id.*

[17] Frederick L. Kirgis Jr., *Aviation*, 2 UNITED NATIONS LEGAL ORDER, 825, 854 (Oscar Schachter & Christopher C. Joyner eds. Cambridge Univ. Press 1995).

transport.[18] As for space law, the issue of unfettered freedom in regard to carrying out exploration and use of space is a pre-eminent concept which prompted one scholar to observe that ninety percent of work in space law is devoted to the problem of sovereignty.[19] The word "freedom", which occurs in both air law and space law, should not be confused with sovereignty. The former relates to freedom of outer space and the latter refers to the freedom of the air which are conferred by sovereign nations upon one another (except for the first and second freedoms which are considered the right to fly over a territory of a State and to land in the territory of one State for non-traffic purposes).[20] Jennings offered the most succinct explanation of freedom in air law when he said:

> The truth of the matter seems to be that the principle of freedom alone is not sufficient to solve the problems of air law; like that of sovereignty, it can serve no more than as a point of departure for the development of an adequate and appropriately elaborate system of rules.[21]

This statement, made when the process of regulation of civil aviation on a global basis was incipient, carries the wisdom of time and is eminently applicable now. Although statements are made conceptually on state sovereignty or freedom of action as being entrenched (such as the one referred to above regarding the 5th ICAO Worldwide Air Transport Conference) in practicality State sovereignty often gives way to regional or community rules and, in some instances, exigencies of economic necessity and compulsion, only to be used as a potential privilege in departing from established rules constituting multilateral agreement. Sovereignty, both in terms of air law and space law, signifies a certain independence of action by a State with regard to

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[18] *Report of the Worldwide Air Transport Conference - Challenges and Opportunities of Liberalization*, U.N. ICAO ATConf/5, 2003, at 59-63, U.N. Doc. 9819 (2003).

[19] Y.A. Korovin, *Conquest of Outer Space and Some Problems of International Relations*, INTERNATIONAL AFFAIRS MOSCOW, 95 (1959) (statement of Eugene Pepin).

[20] *Manual on the Regulation of International Air Transport*, at 4.1.5 and 4.1.8 - 4.1.11, U.N. ICAO Doc 9626 (second ed., 2004).

[21] Robert Y. Jennings, *Some Aspects of the International Law of the Air*, 75 RECUEIL DES COURS 514, 585 (1949).

other States, to their exclusion. However, air law signifies control by a State exercised over its air space whereas no State can exercise sovereignty over any portion of outer space. This prohibition is based on the principle that any projection of territorial sovereignty into space beyond the atmosphere would be inconsistent with astronomical facts which bring to bear the peripatetic nature of the universe. The revolution of the Earth on its own axis, its rotation around the sun and the motion of the sun and planets through the galaxy would effectively preclude any presumptive declaration of territory by an individual state.[22]

The distinction between air space and outer space, particularly in the context of delimiting the two, was not an issue of urgency in 1959 when the United Nations General Assembly Resolution 1348 was adopted. The Ad Hoc Committee on the Peaceful Uses of Outer Space, stated in 1959 that the Committee believed that, in general, the determination of precise limits for airspace and outer space did not present a legal problem calling for priority consideration at that time.[23] Since then there has been a proposal that the upper limit of the "atmosphere" be considered as the extent to which air space would go vertically.[24] This has led to some controversy as the "atmosphere" comprises several layers such as the troposphere, stratosphere, ionosphere and exosphere, all of which have been considered as possible outer limits of airspace.[25] ICAO involvement does not arise if a space object traverses solely the airspace of the launch state. However, as Manfred Lachs points out:

> ...the law has traced no frontier between air space and outer space. Wherever that frontier may eventually be fixed, any object journeying from earth to outer space, or the reverse, must

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[22] C. Wilfred Jenks, *International Law and Activities in Space*, 5 INT'L & COMP. L. Q. 99, 103-04 (1956).

[23] Report to the United Nations General Assembly, *U.N. Ad Hoc Committee on the Peaceful Uses of Outer Space*, 14th Sess., U.N. Doc. A/4141 (1959).

[24] *See* E. Pepin, *Space Penetration, Proceedings*, 52 A.S.I.L. ANNUAL MEETING 229 (1958).

[25] Bin Cheng, *From Air to Space Law*, 13 CURRENT LEGAL PROBS. 228 (1960); *see also*, V.I. LISOVSKY, INTERNATIONAL LAW 196 (Moscow).

needs traverse airspace ... Such problems do arise, however, once the space object enters the airspace of other States.[26]

Dr. Kotaite refers to this possibility when he states that passenger flights will one day operate in an environment congested with space debris. He implies that the day is not in the far distant future when aircraft will take off from one State and traverse outer space before entering the Earth's atmosphere and landing in another State.[27]

III. THE ICAO INVOLVEMENT

Any possible involvement by ICAO in future regulation of outer space activities would be inextricably linked to the nexus between air space and outer space and the blurring of distinction between aircraft *per se* and an aerospace plane.[28] Critical to this merger of activities would be issues involving security and safety, along with commercial and trading issues including market access and competition. ICAO is armed with an excellent model treaty — the Chicago Convention — which has stood the test of time and has already provided proven guidelines for every conceivable aspect of international civil aviation through its 18 Annexes. However, any involvement of ICAO in outer space affairs has to be cautiously considered with a view to finding the best possible manner in which ICAO could effectively contribute to the already existing structure including UNCOPUOS[29] and other international bodies involved in outer space affairs[30] which have so far done a commendable job of de-

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[26] MANFRED LACHS, THE LAW OF OUTER SPACE: AN EXPERIENCE IN CONTEMPORARY LAW MAKING 59 (Sijthoff: Leiden, 1972); *see also*, CARL Q. CHRISTOL, THE INTERNATIONAL LAW AND OUTER SPACE, INTERNATIONAL LAW STUDIES 171 (1962).

[27] Kotaite, *supra* note 1, at 5.

[28] Ruwantissa I. R. Abeyratne, *The Aerospace Plane and its Implications for Commercial Air Traffic Rights*, THE AVIATION Q. 186, 192 (January 1997).

[29] NANDASIRI JASENTULIYANA, A SURVEY OF SPACE LAW AS DEVELOPED BY THE UNITED NATIONS, PERSPECTIVES ON INTERNATIONAL LAW 350-83 (N. Jasentuliyana ed., Kluwer Law International 1995) (explaining and documenting UNCOPUOS and law-making process in outer space affairs); *see also* I.H. PH. DIEDERIKS - VERSCHOOR, AN INTRODUCTION TO SPACE LAW 6 (Kluwer Law International, 2d rev. ed. 1999).

[30] UNCOPUOS is not the only international forum in which legal issues relating to outer space affairs are addressed. The International Telecommunications Union (ITU) is also involved through its regulation of the utilization of the frequency/spectrum re-

veloping material, given the fact that it is a Committee with a technical and legal subcommittee each and lacks the law making power of ICAO.

A good starting point, and one which adds increasing credibility to a possible role to be played by ICAO in outer space affairs, is that there are some who believe that "aerospace law" should be one branch of the law singularly governing aerospace, and have defined "aerospace law" as follows:

> The earth's envelope of air and space above it, the two considered as a single realm for activity in the flight of air vehicles and in the launching, guidance and control of ballistic missiles, earth satellites, dirigible space vehicles and the like.[31]

A pioneer in aerospace law, Dr. John Cobb Cooper, uses the above criteria to further define "aerospace law" "the body of legal principles and rules, from time to time effective, which govern and regulate" aerospace and flight. The space component includes outer space and "its relationship to land and water areas on the surface of the earth, the extent and character of the right of individuals and States to use and control such space, or parts thereof, or celestial bodies therein for flights or other purposes." The air space component includes flight and encompasses "instrumentalities with which flight is effected, including their nationality, ownership, use or control." Also included within the air component of aerospace are "surface facilities in connection with flights, such as airports, other launching or landing areas, navigation facilities and airways." Cooper also considered aerospace to involve the relationship "of every kind affecting or between individuals, communities or states arising from the existence or use of the area of flight (aerospace), or the

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source of the geo-stationary orbit. *See International Space Law in the Making, in* 1 F. FOR AIR AND SPACE L. 3-4 (Marietta Benko & Kai-Uwe Schrogl eds., Frontieres 1993). Additionally, The International Atomic Energy Agency (IAEA) addresses issues concerning environmental problems and the use of nuclear energy resources in outer space. *Id.* The United Nations Educational, Social and Cultural Organization (UNESCO) considers the effects of broadcasting by satellites from a legal, political, social and cultural perspective. *Id.*

[31] DIEDERIKS-VERSCHOOR, *supra* note 29, at 8; John Cobb Cooper, *Aerospace Law - The Subject Matter and Terminology*, 29 J. AIR L. & COM. 89, 89 (1963).

instrumentalities or facilities used in conne[ct]ion therewith or to make the flight effective."[32]

We have reached an age where satellites of various kinds, whether they be for communication, reconnaissance, observation or monitoring, – image remote corners of the Earth and invade the skies. In this sense, the very concept of state sovereignty has been marginalised and the integrity of States violated. Space stations and other space objects such as telescopes carry out numerous activities which may bear upon lower depths of the skies, thus affecting state sovereignty. It is therefore necessary to revisit the direction taken by technological developments in aerospace activities with a view to updating the legislative structure relating to aerospace.

The space shuttle, which was generally defined as a spacecraft, as it was a rocket propelled vehicle designed to move in Earth orbit and outer space, also had the capacity to take off in a manner similar to that of a conventional aircraft. This led to questions being asked regarding what were called "near Earth surface vehicles" which are rocket propelled hybrid space shuttles carrying distinct technical features of an aircraft and performing certain terrestrial functions which is usually carried out by aircraft.[33]

An ICAO-UNCOPUOS synergy which may be established through the ICAO Council and UNCOPUOS, and based on past ICAO work in CNS/ATM systems, warrants some study and cautious scrutiny. ICAO has, throughout the 1990s and through the current century so far, initiated an intense work programme in the technical field of CNS/ATM systems. This has involved sustained work on the development of SARPs for the air ground sub networks of the ATN.[34] In this sense, it would not be unreal-

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[32] Cooper, *supra* note 31, at 94. Dr. Cooper also made the suggestion that at any particular time, the territory of each state extends upward into space as far as the scientific progress of any state in the international community permits such state to control space above it. *Id.* Although this by no means attributes to Dr. Cooper the idea that individual states can claim sovereignty in outer space, this statement brings to bear the relevance of "control" exercised by a state in outer space. *Id.* *See* John Cobb Cooper, *High Attitude Flight and National Sovereignty*, 4 INT'L L. Q. 411 (1951).

[33] S. Mishra & T. Pavlasek, *On the Lack of Physical Bases for Defining a Boundary Between Space and Outer Space*, 7 ANNALS OF AIR & SPACE L. 399, 409 (1982).

[34] Howell, *supra* note 6.

istic to embark on a feasibility study of the development of an Annex to the Chicago Convention with full ICAO-UNCOPUOS involvement and cooperation, on the aerospatial impact on aeronautical activities relating to the safety and security of the aerospace industry and the integrity of the community of states.

The first step therefore would be for ICAO to study the impact of outer space activities of states and private entities on aeronautical activities regulated by ICAO. Such a study could give ICAO and UNCOPUOS an insight as to whether some cooperation between ICAO and other United Nations bodies involved in the regulation of outer space activities is warranted. Article 44 of the Chicago Convention, which sets out the "aims and objectives" of ICAO, identifies them as being, *inter alia*, to "[i]nsure the safe and orderly growth of international civil aviation throughout the world;" "[e]ncourage the arts of aircraft design and operation for peaceful purposes;" and "encourage the development of airways, airports, and air navigation facilities for international civil aviation".[35] These aims and objectives give ICAO ample scope to delve into technological advances made in the design and manufacture of "aerospacecraft" which may, in the future have aeronautical consequences and which may assist in the determination of ICAO's involvement in the regulation of aerospace activities.

Pursuant to Resolution A31-7 adopted by the 31st Session of the ICAO Assembly, in December 1995, the Council of ICAO established a Panel of Experts on the Establishment of a Legal Framework with Regard to Global Navigation Satellite Systems (LTEP) The LTEP was charged, *inter alia*, to develop a legal framework of conduct regarding the use of the Global Navigation Satellite Systems (GNSS). The framework developed by the LTEP took the form of a Charter containing various principles for the implementation and operation of GNSS, such as recognition of the paramount nature of safety in international civil aviation; non-discrimination and universal applicability and accessibility of GNSS; inviolability of States' sovereign rights; continuity, integrity, availability and reliability of ser-

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[35] Chicago Convention, *supra* note 4, at art. 44.

vices; and international cooperation. The Charter was recognized by the ICAO Council, at its 153rd Session in March 1998, as being worthy of submission to the 32nd Session of the Assembly, which in turn adopted the text of the Charter in a Resolution. Therefore, a legal regime already exists in the international arena relating to the provision of satellite communications for air navigation. At its 32nd Session, held in Montreal from 25 September to 5 October 2001, the ICAO Assembly adopted Resolution A32-19, the *Charter on the Rights and Obligations of States Relating to GNSS Services.*[36] The Assembly, while recognizing that Article 44 of the Chicago Convention mandates ICAO to develop the principles and techniques of international air navigation and to foster the planning and development of international air transport, observed in its second preambular clause that the concept of CNS/ATM systems utilizing satellite-based technology was endorsed by States and international organizations at the ICAO Tenth Air Navigation Conference, and was approved by the 29th Session of the Assembly as the ICAO CNS/ATM systems. The Assembly also recognized that the GNSS, was an important element of the CNS/ATM systems, and was intended to provide worldwide coverage and is to be used for aircraft navigation. It was concluded that GNSS would be compatible with international law, including the Chicago Convention, its Annexes and the relevant rules applicable to outer space activities. As such, the Assembly was of the view that it was appropriate, taking into account current State practice, to establish and affirm the fundamental legal principles governing GNSS which should be established in a Charter. The Assembly consequently adopted the principles in the Charter on the Rights and Obligations of States Relating to GNSS Services (GNSS Charter) and which shall apply in the implementation and operation of GNSS.[37]

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[36] *Assembly Resolutions in Force,* ICAO Doc 9790, at V-3 (as of Oct. 5, 2001).

[37] Charter on the Rights and Obligations of States Relating to GNSS Services, 32nd Assembly, ICAO Doc. A32-19 (1998) [hereinafter GNSS Charter].

1. States recognize that in the provision and use of GNSS services, the safety of international civil aviation shall be the paramount principle.

It is at this point that the problem arises, and legal discourse begins. The mere fact that the GNSS Charter is now an ICAO Assembly Resolution has prompted the comment:

> Adopted in the form of an Assembly Resolution, the Charter cannot be accorded any legal force and therefore must be regarded as legally not binding. Some commentators, having expressed serious doubts as to the usefulness of the instrument, seem to be somewhat displeased with the nomenclature em-

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2. Every State and aircraft of all States shall have access, on a non-discriminatory basis under uniform conditions, to the use of GNSS services, including regional augmentation systems for aeronautical use within the area of coverage of such systems.

3. a) Every State preserves its authority and responsibility to control operations of aircraft and to enforce safety and other regulations within its sovereign airspace.

   b) The implementation and operation of GNSS shall neither infringe nor impose restrictions upon States' sovereignty, authority or responsibility in the control of air navigation and the promulgation and enforcement of safety regulations. States' authority shall also be preserved in the coordination and control of communications and in the augmentation, as necessary, of satellite-based air navigation services.

4. Every State providing GNSS services, including signals, or under whose jurisdiction such services are provided, shall ensure the continuity, availability, integrity, accuracy and reliability of such services, including effective arrangements to minimize the operational impact of system malfunctions or failure, and to achieve expeditious service recovery. Such State shall ensure that the services are in accordance with ICAO Standards. States shall provide in due time aeronautical information on any modification of the GNSS services that may affect the provision of the services.

5. States shall cooperate to secure the highest practicable degree of uniformity in the provision and operation of GNSS services. States shall ensure that regional or subregional arrangements are compatible with the principles and rules set out in this Charter and with the global planning and implementation process for GNSS.

6. States recognize that any charges for GNSS services shall be made in accordance with Article 15 of the Chicago Convention.

7. With a view to facilitating global planning and implementation of GNSS, States shall be guided by the principle of cooperation and mutual assistance whether on a bilateral or multilateral basis.

8. Every State shall conduct its GNSS activities with due regard for the interests of other States.

9. Nothing in this Charter shall prevent two or more States from jointly providing GNSS services.

> ployed which would be indicative of a legal instrument of fundamental importance.[38]

This is seemingly consistent with another view on the ICAO Assembly which states that:

> [ICAO] Assembly recommendations ... are more than hortatory. They are designed to set global norms in a field where there is widespread acknowledgment of the need for ordered conduct. They are adopted by a plenary body, with the shared expectation that States will follow them to the extent that they are able to. They clearly are not binding, but they have a sufficient channelling effect to place them well above the low point on a continuum of normative instruments ranging from non law to true law.[39]

However, unlike the former view, this statement attributes more coercive force to ICAO Assembly Resolutions, thus making the GNSS Charter an instrument which sets out norms and demands States to follow them if possible and is therefore in between "non law" and "true law". What this means is that the GNSS Charter would not be totally destitute of effect in establishing certain obligations for States to perform. Therefore, it becomes a reckonable force in international relations, if not at international law, particularly since ICAO resolutions are highly persuasive and carry much political leverage. Above all, such an instrument could, while reaffirming existing legal principles, pave the way for an international convention that is binding on States' Parties.

The significance of the legal status regarding the current principles on the conduct of States in using space based applications in air traffic management lies in the compelling need to inquire as to whether rigid fragmentation of law and policy is really necessary, particularly in such an important area as aviation safety. Does one dismiss policy in this critical area purely on the inflexible notion that it is not enforceable? On the other hand, do States need to abandon rigid demarcations in in-

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[38] ALESSANDRA A.L. ANDRADE, THE GLOBAL NAVIGATION SATELLITE SYSTEM: NAVIGATING INTO THE NEW MILLENNIUM 89 (Ashgate 2001).

[39] Kirgis *supra* note 17, at 840.

stances such as these and agree to global adherence? These are some of the issues that have to be ironed out when the subject of cooperation between the aviation and space communities surfaces.

Also at the 32nd Session, the Assembly adopted Resolution A 32-20, Development and Elaboration of an Appropriate Long-term Legal Framework to Govern the Implementation of GNSS.[40] It was recognized that the GNSS was an important element of the ICAO CNS/ATM systems, is intended to provide safety-critical services for aircraft navigation with world-wide coverage, and that the complex legal aspects of the implementation of CNS/ATM, including GNSS, require further work by ICAO. Further work is necessary in order to develop and build mutual confidence among States regarding CNS/ATM systems and to support the implementation of CNS/ATM systems by Contracting States.

The Assembly recalled that the world-wide CNS/ATM systems implementation Conference in Rio de Janeiro in May 1998 recommended that a long-term legal framework for GNSS be elaborated, including the consideration of an international convention, while recognizing that regional developments may contribute to the development of such a legal framework and that the recommendations adopted by the Rio Conference as well as the recommendations formulated by the LTEP provide important guidance for the development and implementation of a global legal framework for CNS/ATM and in particular GNSS. Consequently, the Assembly, while recognizing the importance of regional initiatives regarding the development of the legal and institutional aspects of GNSS, and the urgent need for the elaboration, both at a regional and global level, of the basic legal principles that should govern the provision of GNSS, called for an appropriate long-term legal framework to govern the implementation of GNSS. Recalling an earlier decision of the Council taken on 10 June 1998 authorizing the Secretary General to establish a Study Group on Legal Aspects of CNS/ATM systems, the Assembly instructed the Council and the Secretary General,

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[40] *Id.* at V-4.

within their respective competencies, and beginning with a Secretariat Study Group, to ensure the expeditious follow-up of the recommendations of the world-wide CNS/ATM Systems Implementation Conference, as well as those formulated by the LTEP. Special consideration was to be given to those recommendations concerning institutional issues and questions of liability. Elaboration of an appropriate long-term legal framework to govern the operation of GNSS systems, including consideration of an international Convention for this purpose was also to be considered. Presentation of proposals for such a framework was to be made in time for consideration by the next ordinary Session of the Assembly.

IV. CONCLUSION

ICAO's involvement in outer space activities would hinge on the nature of technological advancements to come regarding the design and manufacture of spacecraft and their use of air space and outer space. The former is relevant since ICAO has an obligation to the international community, as mentioned above, under Article 44 of the Chicago Convention. The latter is important as the skies are becoming seamless and the single skies concept, particularly in Europe, is gaining on the aviation community. The use of the sky, from an air navigation service provider perspective, as well as from a territorial perspective, has become a serious consideration making it necessary for the legal ramifications of the issue of airspace to be given attention.

Should the consequences of future outer space activities have a serious effect on ICAO's aeronautical activities, they would have to be studied in depth and appropriate synergies between the aviation and outer space developed. What is immediately required is a vision both for the aviation community and outer space affairs community, for effective cooperation within their competencies, aims and objectives

It should not be forgotten that another positive feature of ICAO, which lends itself well to the Organization's possible involvement in outer space activities, is its proven competence in mediation during the settlement of disputes. A preeminent feature of the ICAO Council is its indomitable resolve to address its

deliberations to purely technical issues pertaining to any dispute, while stringently avoiding political issues and pitfalls. This is certainly true of all disputes brought before the Council, where the Council restricted its scope to technical issues as applicable to the principles embodied in the Chicago Convention.

Also to be remembered is that public international law is increasingly becoming different from what it was a few decades ago. We no longer think of this area of the law as a set of fixed rules, even if such rules have always been a snapshot of the law as it stands at a given moment. Fundamentally, and at its core, international law was considered in simple terms as the law binding upon States in their relations with one another.[41] The abovementioned principle was implicitly derived from the basic rule of law as it applies even today, that in the sustained evolution of humanity from troglodytes to computer wizards a central role has always been played by the idea of law – the idea that in every civilized society there must be order as against chaos and anarchy which were inimical to a just and stable society. Therefore law is the glue which binds the members of a community, whether national or international, together in their adherence to recognized values and standards. In international law,[42] the principal subjects are nation States, not individual citizens. Public international law applies to relations between and among States in all their numerous and complex forms, from war to satellites, and governs operational policy of many international institutions. Some of the new and emergent areas of international law govern: the use of radio frequencies; communications; the availability, exploration and exploitation of resources, whether in the sea bed or in outer space; multinational corporations; trade, investment and finance; pollution, in all its forms; international crime and multinational corporations.[43]

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[41] Robert Y. Jennings, *An International Lawyer Takes Stock*, 39 INT'L & COMP. L. Q. 513 (1990).

[42] International law itself is divided into private and public international law, the former being also referred to as "conflict of laws" and the latter just termed "international law". MALCOLM N. SHAW, INTERNATIONAL LAW 1 (Cambridge Univ. Press: 5th ed., 2003).

[43] Jennings, *supra* note 41, at 521.

International law and politics overlap in instances where international disputes may emerge between or among nations. International law has no legislature. Although the General Assembly of the United Nations exists and functions as a regulator of international policy, being composed of delegates from all member States of the United Nations, its resolutions are generally not binding on member States,[44] except in certain circumstances. The United Nations system has no system of courts except for the International Court of Justice which can only hear cases between States if both sides to a dispute agree.[45] Even if the parties to a dispute agree to come before the Court, it has no jurisdiction to make sure that its decision is enforced or followed. In such an environment, an organization such as ICAO, with its strong mediatory history could prove to be invaluable.

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[44] *See* U.N. CHARTER arts. 10 and 11. (alluding to General Assembly making recommendations to member States); *see also*, D. H. N. Johnson, *The Effect of the Resolutions of the General Assembly of the United Nations*, 32 BRIT. Y. B. INT'L. L. 97 (1955).

[45] *See* The Statute of the International Court of Justice, June 26, 1945, art. 36 (2), 59 Stat. 1031 (entered into force Oct. 24, 1945) (calling for States Parties to Statute to declare consensually that they recognize jurisdiction of Court).